

Attendance Management System

Group 05

Rajinthan R.	220502M
Praveen V.V.J.	220491B
Sulojan R.	220626V



How System Works

- Automate the process of taking attendance in live physical classes.
- A handheld **NFC reader** records each student card and the time.

01

Pass the Reader



02

Tap the Student ID



03

Instant Feedback and
Verification

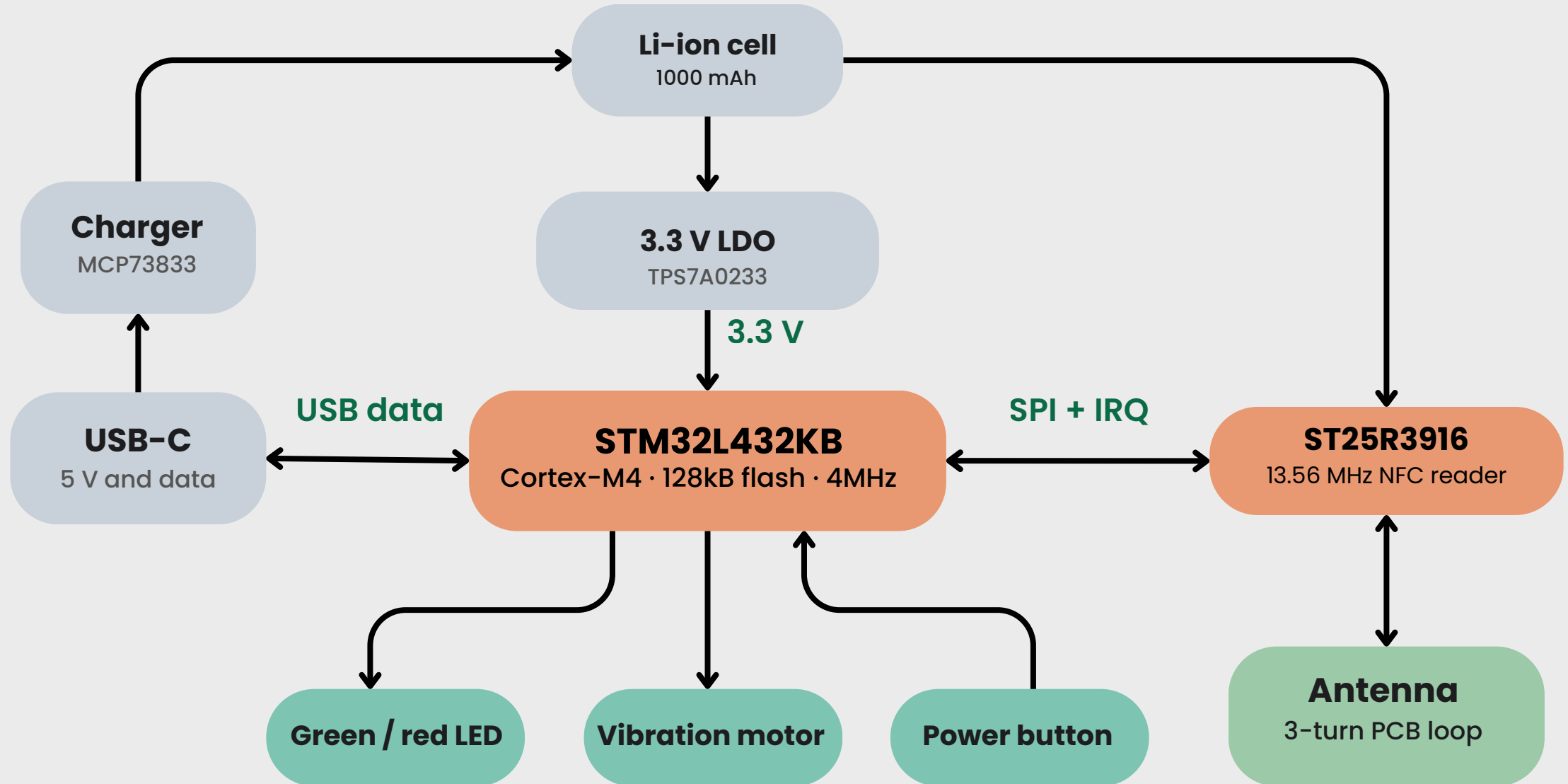


04

Plug in Over USB to
retrieve records



Architecture



Power Analysis

1.8 μA

Switched off (Standby)

108 μA

Idle and waiting for a card

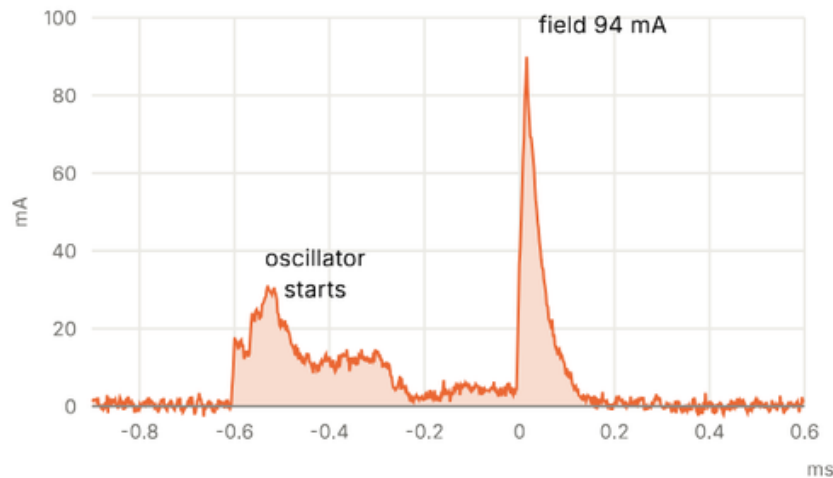
2.8 μAh

One card tap (10.3 mC)

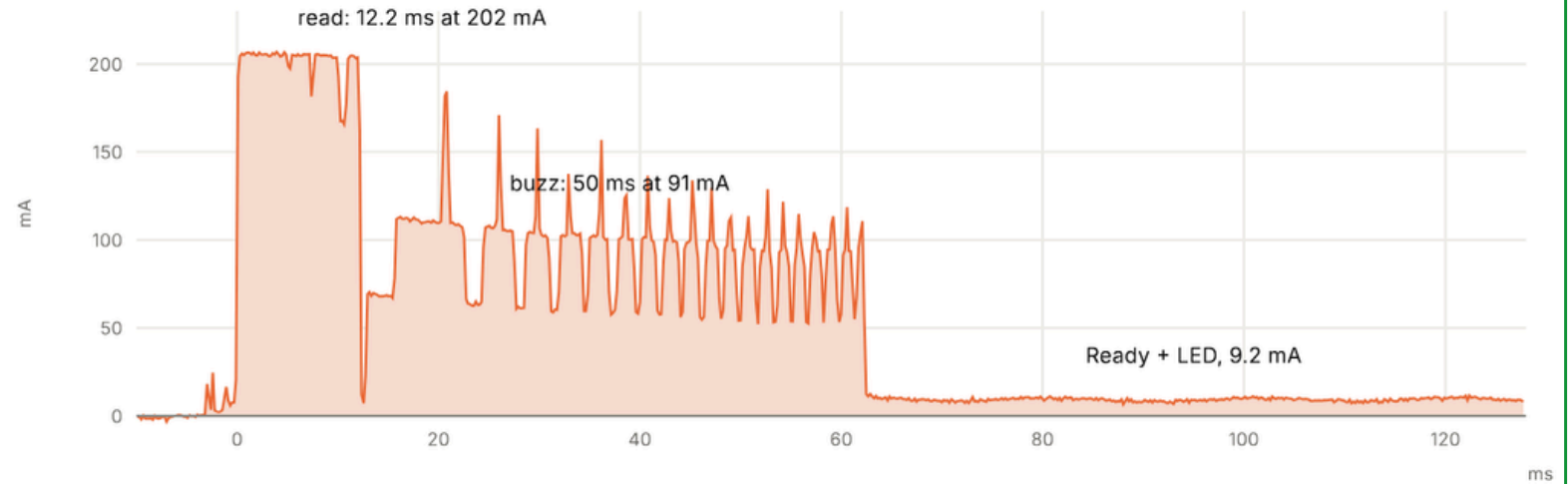
17 months

On a 1000 mAh cell
(12 worst case)

One wake-up measurement (TEK00005)

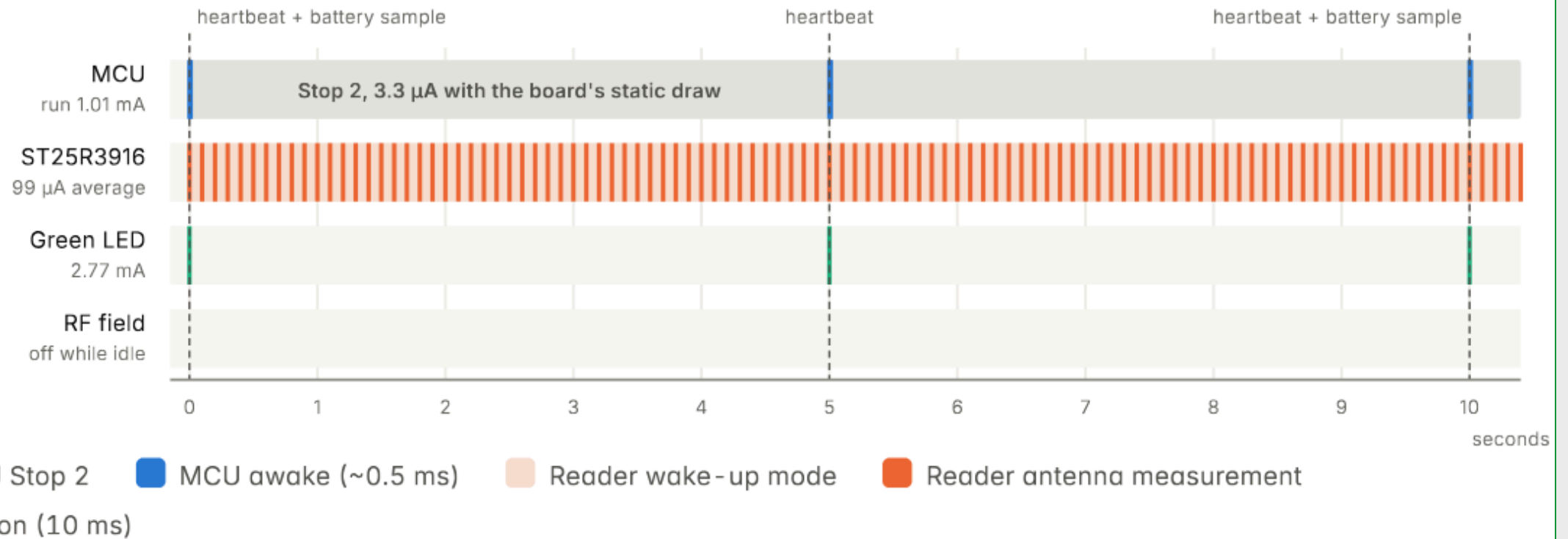


A card tap (TEK00006)



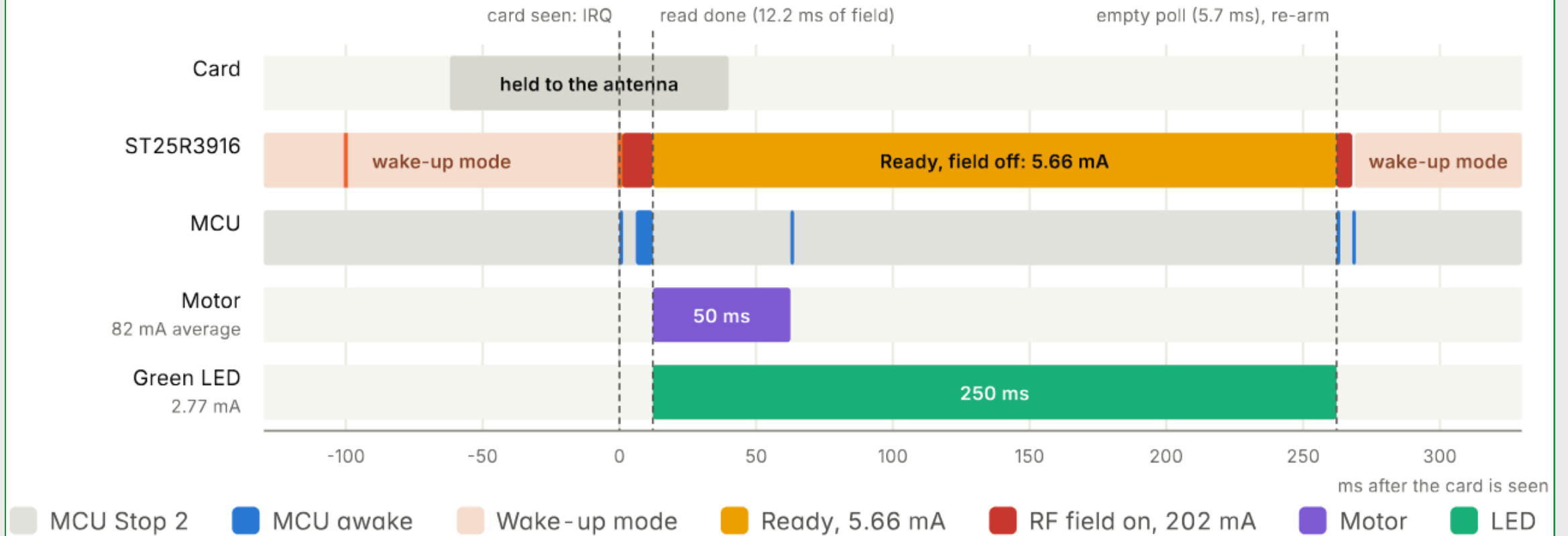
Idle Scanning

Idle scanning: ten seconds



One card Tap

One accepted card tap



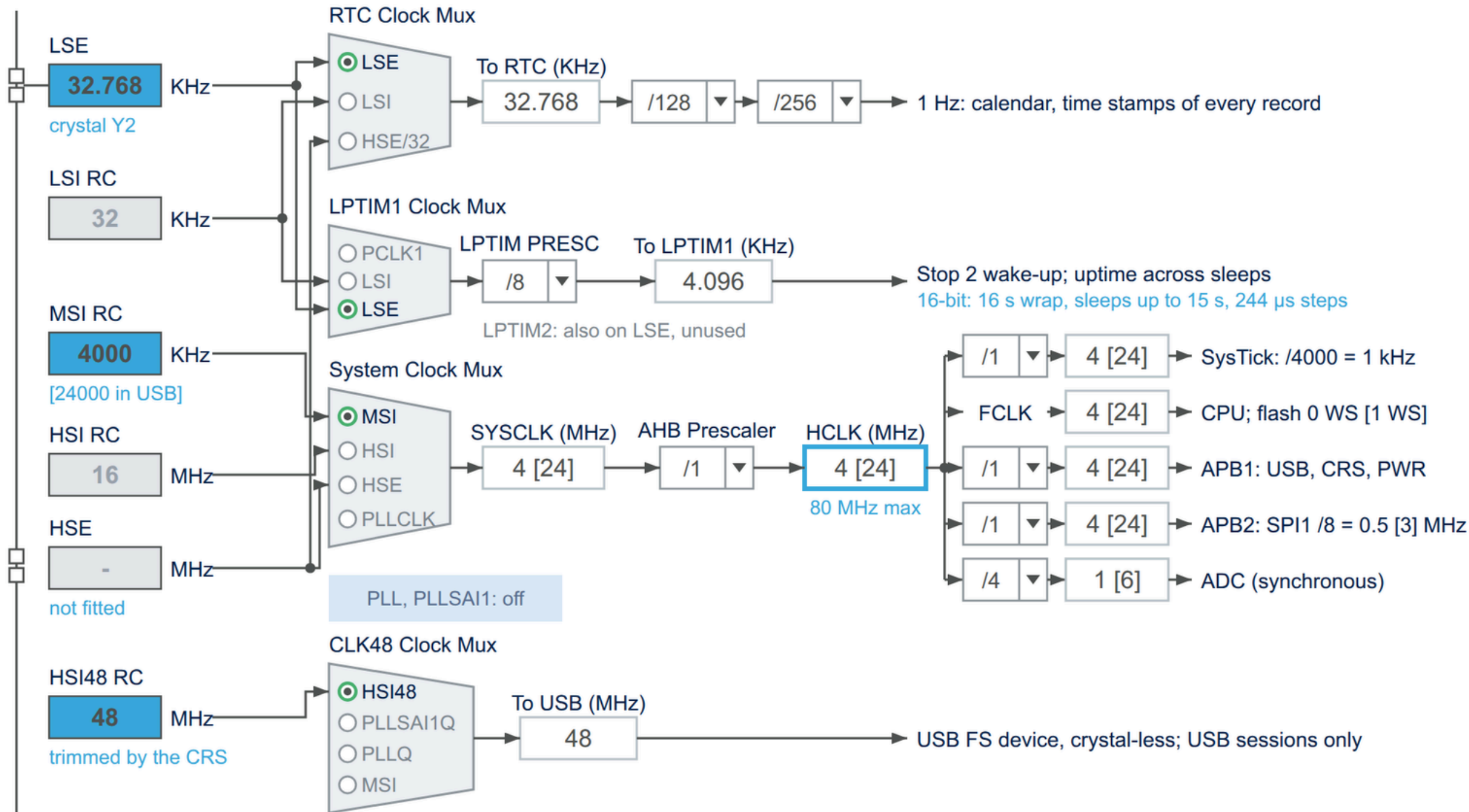
Clock Speed Calculation

Clock	Speed	Why
Scanning	4 MHz (MSI)	no PLL to relock on each wake-up
USB session	24 MHz (MSI)	USB needs above 14.2 MHz; power comes from the cable
USB clock	HSI48 + CRS	locks to the host, no crystal
RTC, timers	32.768 kHz	runs through sleep for under 1 μ A
Reader SPI	500 kHz	chip allows 10 MHz

Charge in one 15 ms card read

MCU at 4 MHz (chosen)		15.2 μ C
MCU at 24 MHz		42 μ C
MCU at 80 MHz (PLL)		135 μ C
RF field for the same read		2030 μ C

Reading a card waits on the radio, not on the processor, so **the slowest clock that meets the timing wins**. At 4 MHz the MCU is under 1 % of a read.



Memory

STM32L432KB: **128kB** of flash, 64 pages of 2 kB

Program: 36 pages, 72 kB

image 60.6 kB at -Og, 11.4 kB free

Attendance log: 27 pages

6,858 records

● program ● config page (device ID) ● log

- One record is **8 bytes**: card number and time. That is exactly one flash double-word.
- A page holds 254 records. **6,858 records** is about **64 lectures** of 100 students.
- The log fills in about 16 teaching days. The app clears it each time a lecture starts.

Build	Image	Fits 72 kB
Debug (-Og)	62.1 kB	yes, 11.4 kB free
Release (-Os)	51.5 kB	yes, 21.7 kB free

Justification of NFC

- Student ID cards and NFC tags are usually **13.56 MHz**. One reader IC, the **ST25R3916**, reads them over SPI.
- Every card has its own number (4, 7 or 10 bytes). That number is all the device stores.
- The reader can **look for a card by itself** every 100 ms while the MCU sleeps.
- The antenna is a **3-turn loop etched on the PCB**: no extra part.

	First design 125 kHz EM4100	Final 13.56 MHz NFC
Cards	EM4100 tags	ISO 14443-A cards
Reader	discrete circuit	one chip

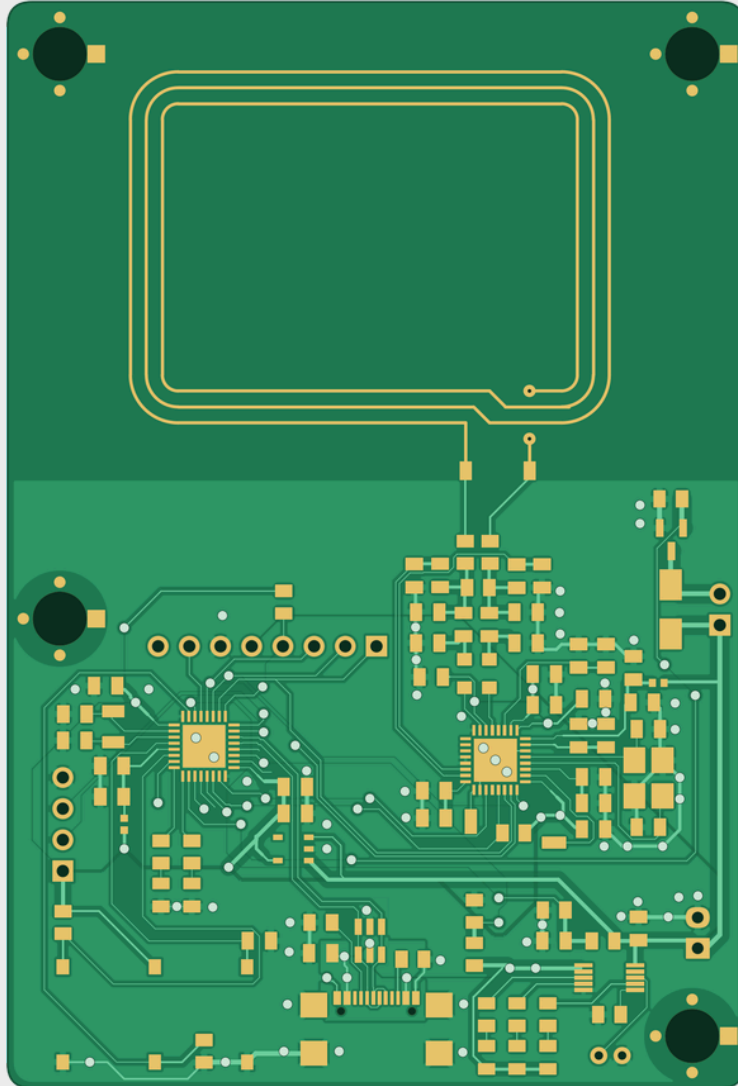
18.2_mA

polling the reader all
the time(125 kHz)

194_μA

with the reader's
wake-up mode

PCB Design



- loop antenna etched on the board, 3 turns.
- USB-C at the bottom edge charges the cell and shows the drive.
- Reader and motor run straight from the cell; the MCU from a 3.3 V LDO.

Block	Part
MCU	STM32L432KB
NFC reader	ST25R3916
Charger	MCP73833
LDO	TPS7A0233
USB protection	USBLC6-2SC6

The Companion App

The screenshot shows the 'Attendance Logger' app interface. At the top, there's a menu bar with 'File', 'Device', 'View', and 'Help'. Below it, a header bar displays 'Attendance Logger' with a checkmark icon, 'RFID attendance for your lectures', a battery status of 87%, and a notification 'Demo device · 151 taps on the device'. A toolbar contains buttons for 'Connect', 'Read device', 'Eject', 'Set device time', and 'Clear device records'. The main area is divided into a sidebar with 'Lectures' and 'Students' options, and a central content area. The 'Start a lecture' section has a text input 'Circuits Lecture 5' and a 'Start lecture' button. Below it, a message explains how to use the device. The 'Lectures' section shows a list of 8 lectures with columns for Lecture, Date, Start, and Present. The 'Circuits Lecture 4' section shows the current lecture status (Thu 08 Oct 2026, started 19:05, still running) and a '17 present' count. It includes a search bar and a table of students with columns for Index No, Name, and Time. A message indicates '2 cards tapped in this lecture but are not registered yet.' with a 'Register...' button. At the bottom, there are buttons for 'Rename...', 'End lecture', 'Delete', 'Save list (CSV)...', and 'Save as PDF...'. A footer note states 'Demo data: nothing here is real · 30 students · 8 lectures · 151 taps'.

File Device View Help

✓ **Attendance Logger**
RFID attendance for your lectures

87 % • Demo device · 151 taps on the device

Connect Read device Eject Set device time The device clock is right. Clear device records

Lectures

Students

Start a lecture

Circuits Lecture 5 ▶ Start lecture

Saves the device's taps here, sets its clock and starts the lecture. No computer at hand? Hold the device's button until it buzzes (2 seconds) and let go: the next lecture starts, and appears here the next time you plug the device in.

Lectures 8 lectures Clear all records...

Lecture	Date	Start	Present
• Circui...	2026-10-08	19:05	17
Circuits L...	2026-10-01	09:00	15
Calculus ...	2026-09-25	09:00	25
Circuits L...	2026-09-24	09:00	17
Intro to ...	2026-09-23	11:00	9
Intro to ...	2026-09-23	09:00	10
Calculus ...	2026-09-18	09:00	28
Circuits L...	2026-09-17	09:00	19

Circuits Lecture 4

Thu 08 Oct 2026 · started 19:05 · still running

17 present

Search by name or index number

Index No	Name	Time
EN/21/110	Pradeep Fernando	19:06
EN/21/104	Ruwan Bandara	19:06

2 cards tapped in this lecture but are not registered yet. Register...

Rename... End lecture Delete Save list (CSV)... Save as PDF...

Demo data: nothing here is real · 30 students · 8 lectures · 151 taps

- Plug the device in and the app reads every tap.
- See who came: index number, name, time.
- Save each lecture as **CSV** or **PDF**.
- Register new cards from the device.

Total Cost

Units made	1	100	1,000	10,000
Cost per unit (USD)	227.62	32.43	25.40	18.31
Cost per unit (LKR)	75,115	10,702	8,382	6,042

Where \$25.40 goes (1,000 units)

Electronic parts



\$11.20

Battery, motor, case



\$6.70

Import duty, VAT, levies



\$4.80

PCB and assembly



\$0.96

Test, freight, yield loss



\$1.74

Quick Guide

You see	It means
● Green + one short buzz	Card recorded
● Two quick buzzes	Already recorded in this lecture
● Green + three quick buzzes	New lecture started
● Red blinking 5 times	Battery low
● Red + four buzzes	Memory full, tap not recorded
● Red + long buzz	Switching off

The power button

Press (off)	Switch on
Tap	Show the battery
Hold 2 s	Start next lecture
Hold 5 s	Switch off

It switches itself off after 3 minutes without a tap. Records are kept.

1 Switch on

2 Tap cards

3 Hold 2 s for the next lecture

4 Plug in, open the app